JORDAN LEVY

Applied Scientist ⋄ PhD Candidate in AI

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PRESENTATION

Profile

As an Applied Scientist and third-year PhD candidate (CIFRE), I bridge the gap between scientific AI research and industrial deployment. Splitting my time between Université Toulouse Capitole (IRIT) and TwinswHeel, I combine rigorous research methodology with robust software engineering to design state-of-the-art time-series models and foundation model adaptation for mission-critical autonomous systems. I am actively seeking an Applied Scientist / Research Engineer position in the industry starting in 2027 to develop next-generation time-series and deep learning architectures.

Research Interests

My research interests center on **time-series modeling**, **foundation models**, and **deep learning**, with a core focus on unsupervised anomaly detection, classification, and surrogate modeling. I am dedicated to developing scalable, high-performance architectures capable of continuous online adaptation in demanding real-world industrial environments.

RESEARCH EXPERIENCE

Applied Scientist at TwinswHeel and PhD Candidate *Feb 2024 - March 2027*

University: Université Toulouse Capitole, France
Laboratory: Institut de Recherche en Informatique de Toulouse (IRIT)
Research Team: Cooperative Multi-Agents Systems (SMAC)
Partner Company: TwinswHeel
Supervisors: Benoit Gaudou, Nicolas Verstaevel, Vincent Talon

Keywords: Time-Series Foundation Models ⋄ Anomaly Detection ⋄ Distillation ⋄ Edge AI ⋄ XAI

Summary: Designed and deployed robust time-series anomaly detection architectures for autonomous mobile robot fleets. Investigated foundation model adaptation by distilling deep time-series models (TSPulse) into lightweight, adaptive edge models achieving low-latency onboard inference. Pioneered surrogate modeling and SHAP-based explainability to analyze detector complementarity and ensure reliability under real-world domain shifts.

Research Internship in Trajectory Prediction *March 2023 - Aug 2023*

Company: Continental Digital Services France
Team: Maps on Demand
Supervisors: Anaïs Goursolle, Christophe Ailleres

Keywords: Spatio-Temporal Data - Trajectory Prediction - Deep Learning - PyTorch

Summary: Designed and implemented deep learning models to predict autonomous vehicle trajectories at complex intersections. Developed spatio-temporal architectures (MLP, LSTM, and a custom VectorNet implementation) leveraging GNSS data and HD Maps context. Built and deployed an optimized real-time trajectory prediction pipeline on embedded hardware using Python and PyTorch.

Research Internship in Opportunistic Software Composition *May 2022 - July 2022*

University: Université de Toulouse, France
Laboratory: Institut de Recherche en Informatique de Toulouse (IRIT)

Research Team: Cooperative Multi-Agents Systems (SMAC)
Supervisors: Maxence Demougeot, Kevin Delcourt, Jean-Paul Arcangeli

Keywords: Multi-Agent Systems - Reinforcement Learning - IoT - Java

Summary: Tested and enhanced a reinforcement learning-based multi-agent system for dynamic IoT software composition. Engineered new features for Java/JavaFX simulation tools and established Continuous Integration (CI) pipelines on GitLab.

EDUCATION

PhD in Computer Science *Feb 2024 - March 2027*

Université Toulouse Capitole, France

Engineering Degree in Artificial Intelligence *2020 - 2023*

Polytech Nantes, France

University Exchange Program *2022*

Kyungpook National University, Daegu, South Korea

University Diploma in Technology (DUT) in Computer Science *2018 - 2020*

IUT Paul Sabatier, Toulouse, France

PUBLICATIONS

International Publication

TimeCIEL : Contextual Interactive Ensemble Learning for Time Series Classification, Jordan Levy, Clément Blanco-Volle, Nicolas Verstaevel, Benoit Gaudou, Vincent Talon, International Conference on Practical applications of Agents and Multi-Agent Systems, 2025

Analyzing Shapley Additive Explanations to Understand Anomaly Detection Algorithm Behaviors and Their Complementarity, Jordan Levy, Paul Saves, Moncef Garouani, Nicolas Verstaevel, Benoit Gaudou, IDA 2026 - **Best Paper Award**

Preprints

Continuous Online Fault Detection for Mobile Robots via Adaptive Edge Models, Jordan Levy, Nicolas Verstaevel, Vincent Talon, Benoit Gaudou, arXiv:2609.29194, 2026

Surrogate Modeling and Explainable Artificial Intelligence for Complex Systems: A Workflow for Automated Simulation Exploration, Paul Saves, Pramudita Satria Palar, Muhammad Daffa Robani, Nicolas Verstaevel, Moncef Garouani, Jordan Levy, Julien Aligon, Benoit Gaudou, Koji Shimoyama, Joseph Morlier, arXiv:2510.16742, 2025

SUPERVISED INTERNSHIPS

Mathieu Galinier *Apr 2025 - Sept 2025*

Level: Master's Internship
Project Title: Data Collection and Benchmarking for Anomaly Detection in Autonomous Robotics

Alexandre Maignan *March 2024 - Aug 2024*

Level: Master's Internship
Project Title: Explainable Anomaly Detection for Mobile Robotics

OTHER RESPONSIBILITIES

Reviewer: ICRA 2027, PAAMS 2026
Doctoral Student Committee Member: Representing PhD students in laboratory councils, organizing social events, and facilitating communication with the administration.

PERSONAL PROJECTS & VOLUNTEERING

Kubesleeper *2025*

Co-creator and core developer of the open-source project Kubesleeper (GitHub): a scale-to-zero Kubernetes manager. It automatically reduces resource usage based on load to cut infrastructure costs. Built from scratch using **Rust** and **Kubernetes**.

Google Hash Code *2022*

Team programming competition. Global ranking: 1316/10000.

Lyon e-Sport Volunteer *Feb 2019*

Volunteered as a stage technician for the major e-sports event in Lyon, France.

TECHNICAL SKILLS

Programming:	Python (PyTorch, NumPy, SciPy, Pandas), Rust, C/C++, Java, Bash
Time Series & ML:	Time-Series Foundation Models, Anomaly Detection, Classification, Surrogate Modeling, Explainable AI
DevOps & Systems:	Linux, Docker, Kubernetes, Git, CI/CD, Slurm / HPC, ROS

LANGUAGES

- **French:** Native
- **English:** Advanced
- **Spanish:** Intermediate
- **Korean:** Beginner

HOBBIES

- Cinema
- E-Sport
- Improvisational Theatre
- Board Games